



SYDNEY GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT
TRIAL EXAMINATIONS 2004

FORM VI

MATHEMATICS

Examination date

Wednesday 4th August 2004

Time allowed

3 hours (plus 5 minutes reading time)

Instructions

- All ten questions may be attempted.
- All ten questions are of equal value.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Collection

- Write your candidate number clearly on each booklet.
- Hand in the ten questions in a single well-ordered pile.
- Hand in a booklet for each question, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Keep the printed examination paper and bring it to your next Mathematics lesson.

Checklist

- SGS booklets: 10 per boy. A total of 1250 booklets should be sufficient.
- Candidature: 109 boys.

Examiner

PKH

QUESTION ONE (12 marks) Use a separate writing booklet.

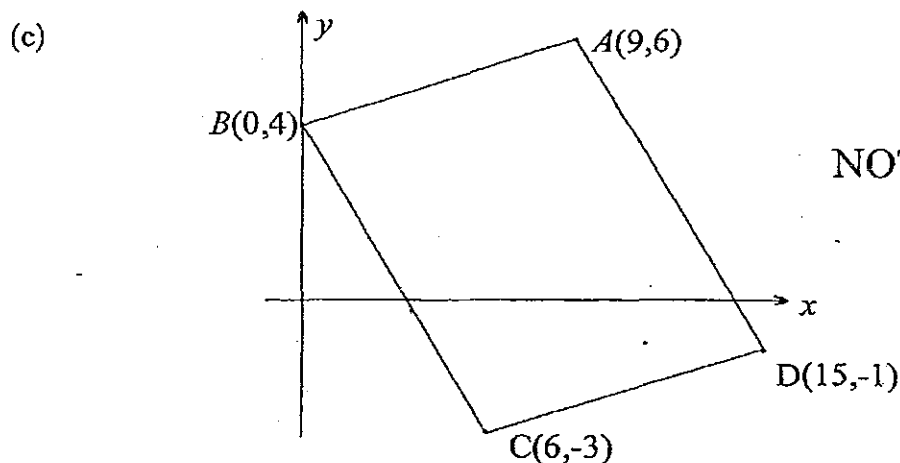
Marks

- (a) Evaluate $\sqrt{\frac{3 \cdot 4^4}{15.6 \times 12.8}}$, correct to three significant figures. 2
- (b) Differentiate $5x^2 - \cos x$. 2
- (c) Find a primitive of $x^3 + 5$. 2
- (d) A sector of a circle subtends an angle of 40° at the centre of the circle. If the radius of the circle is 9 units, find the area of the sector. 2
- (e) (i) Solve $|x - 4| \leq 1$. 1
 (ii) Graph your solution on a number line. 1
- (f) Express $\frac{3}{3 - \sqrt{5}}$ with a rational denominator. 2

QUESTION TWO (12 marks) Use a separate writing booklet.

Marks

- (a) Arthur invests \$20 000 in a term deposit at 6% pa compounded annually. How much will the investment be worth after four years? Give your answer correct to the nearest cent. 2
- (b) Find the equation of the tangent to the curve $y = e^{2x}$ at the point where $x = 0$. 3



In the diagram above, $ABCD$ is a quadrilateral:

- (i) Find the midpoint of the diagonal AC . 1
- (ii) Find the midpoint of the diagonal BD . 1
- (iii) Find the gradient of BD . 1
- (iv) Show that AC is perpendicular to BD . 2
- (v) What shape best describes quadrilateral $ABCD$? Give reasons. 2

Exam continues next page ...

QUESTION THREE (12 marks) Use a separate writing booklet.

Marks

(a) Differentiate the following functions:

(i) $y = \cos(2x + 1)$

2

(ii) $y = \frac{x}{\log_e x}$

2

(iii) $y = x^2 \tan x$

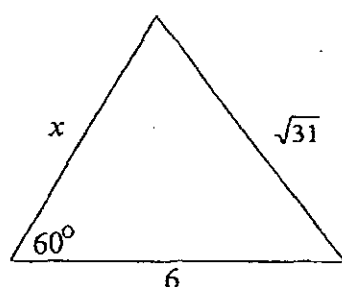
2

(b) Sketch the parabola $x^2 = -4(y + 2)$, showing its vertex and directrix.

3

(c)

3



In the diagram above, use the cosine rule to find the possible values of x .

QUESTION FOUR (12 marks) Use a separate writing booklet.

Marks

(a) Find:

(i) $\int_0^{\ln 2} e^{2x} dx$

2

(ii) $\int \frac{2x}{x^2 + 5} dx$

1

(b) Consider the arithmetic series

$$5 + 12 + 19 + \dots + 292.$$

(i) How many terms in the series?

1

(ii) Find the sum of the series.

2

(c) A particle moves in a straight line. At time t seconds, its displacement x metres from the origin is given by

$$x = 8t - 2t^2.$$

(i) Sketch the graph of x as a function of t , showing the vertex.

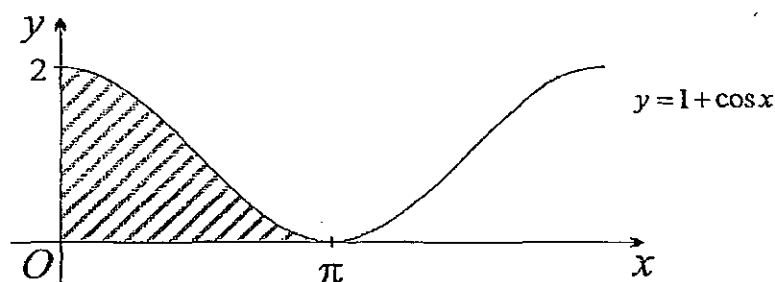
2

(ii) Find the distance the particle travels in the first three seconds.

2

Exam continues overleaf ...

(d)



2

Find the shaded area in the diagram above.

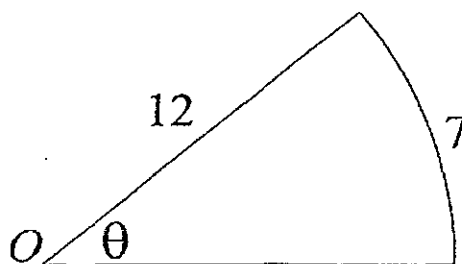
QUESTION FIVE (12 marks) Use a separate writing booklet.

Marks

(a) Sketch the function $y = -2 \sin x$, for $0 \leq x \leq 2\pi$.

2

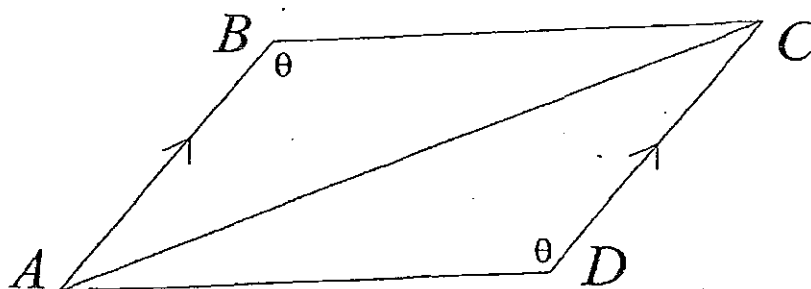
(b)



2

The sector drawn above has its centre at O . Find the size of angle θ , correct to the nearest degree.

(c)



In the diagram above, $AB \parallel DC$ and $\angle B = \angle D = \theta$.

(i) Prove that $\triangle ABC \equiv \triangle CDB$.

2

(ii) Prove that the quadrilateral $ABCD$ is a parallelogram.

3

(d) (i) Make x^2 the subject of $y = \sqrt{x-1}$.

1

(ii) Find the volume formed when the region between the curve $y = \sqrt{x-1}$ and the y -axis, from $y = 0$ to $y = 3$, is rotated about the y -axis.

2

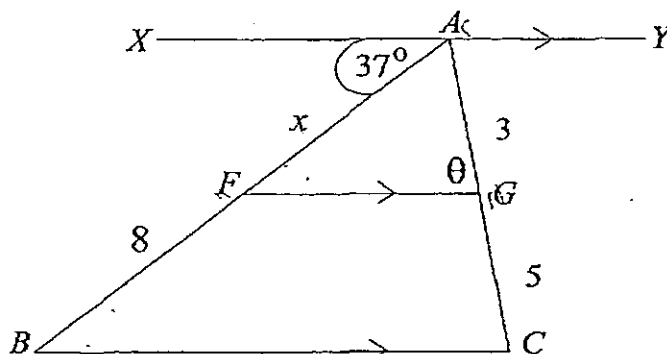
Exam continues next page ...

QUESTION SIX (12 marks) Use a separate writing booklet.

Marks

- (a) The derivative of a function is given by $\frac{dy}{dx} = \frac{1}{\sqrt{x}}$. Given that $y = -2$ when $x = 4$, 2
find y as a function of x .

(b)



- (i) Find the value of x in the diagram above. 2
 (ii) Find θ , correct to the nearest degree. 2
 (c) If $\log_a 5 = x$ and $\log_a 2 = y$, find $\log_a 400$ in terms of x and y . 2
 (d) (i) Copy and complete the table below for $y = \sqrt{2 + e^x}$, calculating each value correct to three decimal places. 2

x	0	1	2
y			

- (ii) Use Simpson's rule with three function values to approximate $\int_0^1 \sqrt{2 + e^x} dx$. 2
 Give your answer correct to two decimal places.

QUESTION SEVEN (12 marks) Use a separate writing booklet.

Marks

- (a) Consider the function $y = x^4 - 4x^3 + 3$.
 (i) Find the stationary points and determine their nature. 4
 (ii) The curve has a point of inflexion where the tangent is not horizontal. Find the coordinates of this point. 2
 (iii) Sketch the function, showing all stationary points and points of inflexion. 2
 (b) For what values of k is the quadratic 4

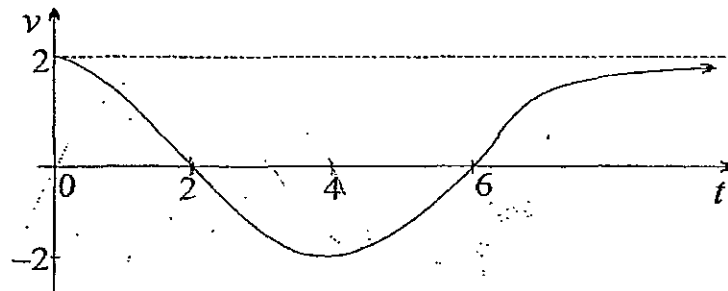
$$kx^2 - 2x\sqrt{6} + k + 1$$
 positive definite?

Exam continues overleaf ...

QUESTION EIGHT (12 marks) Use a separate writing booklet.

Marks

(a)



A particle is moving in a straight line with displacement measured from the origin. The graph drawn above shows the particle's velocity at time t . The particle is initially at the origin.

(i) When does the particle return to the origin?

1

(ii) Draw a graph of the displacement x against the time t .

2

(b) The population P of a growing town satisfies the equation

$$P = P_0 e^{kt}$$

where t is time in years.

The initial population is 22 000 and five years later the population is 27 000.

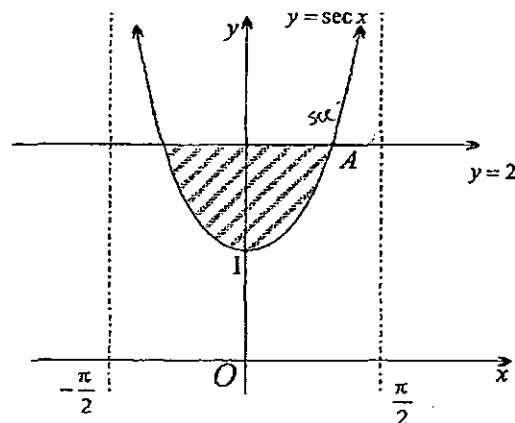
(i) Find P_0 and k .

3

(ii) When does the population reach 35 000? Give your answer correct to three significant figures.

2

(c)



(i) Find the x coordinate of point A in the diagram above.

1

(ii) Find the volume formed when the shaded area in the diagram above is rotated about the x axis.

3

Exam continues next page ...

QUESTION NINE (12 marks) Use a separate writing booklet.

Marks

- (a) A vessel initially contains 100 litres. It is being emptied, and the rate of change of volume is

$$\frac{dV}{dt} = - \left(2 + \frac{20}{t+1} \right)$$

where V is the volume in the vessel in litres after t minutes.

- (i) What is the initial rate $\frac{dV}{dt}$?

1

- (ii) Find how many litres remain in the vessel after five minutes.

3

- (b) A person borrows \$250 000 from a bank at a reducible interest rate of 6% per annum, compounded monthly. The loan is to be repaid in equal monthly installments.

Let $\$M$ be the monthly payment. Let A_n be the amount owing at the end n months when the n th payment has just been made.

The loan must be paid off after twenty years.

- (i) Show that the amount owing after three months is given by

2

$$A_3 = 250\,000 \times (1.005)^3 - M(1 + 1.005 + 1.005^2).$$

- (ii) Explain why $A_{240} = 0$.

1

- (iii) Find the value of M .

3

- (iv) Suppose now that the person elects to pay \$2000 per month instead of the amount calculated in part (iii). How much more quickly would the person pay off the loan?

2

Exam continues overleaf ...

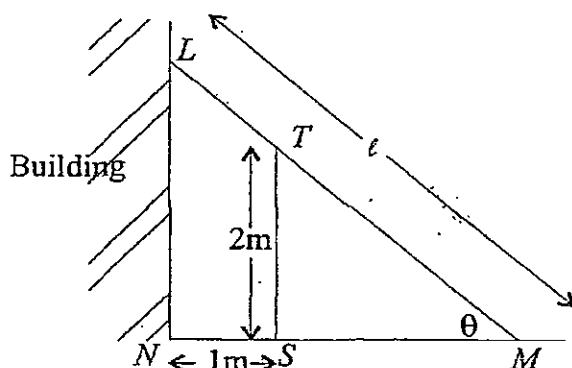
QUESTION TEN (12 marks) Use a separate writing booklet.

Marks

- (a) (i) Show that the graph of $y = x^{\frac{2}{3}}$ is concave down for all values of x except for $x = 0$. 2

- (ii) Solve $x^{\frac{2}{3}} \leq \frac{x}{2}$. 2

(b)



The diagram above shows an extension ladder LM of variable length ℓ . The ladder leans against the wall of a building. It also touches the top of the fence ST , which is 2 metres high and stands 1 metre from the wall.

Let θ be the angle between the ladder and the horizontal ground.

- (i) Show that $\ell = \frac{2}{\sin \theta} + \frac{1}{\cos \theta}$. 2

- (ii) Show that the stationary point of the graph of ℓ occurs when $\tan \theta = \sqrt[3]{2}$. 3

- (iii) For safety reasons, θ must lie between 55° and 70° . Find the minimum length of ℓ . Justify your answer. 3

END OF EXAMINATION

SOLUTION TO SGS TRIAL 2004 Mathematics

Question One (12 marks)

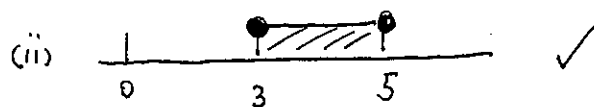
(a) $\sqrt{\frac{3.4^4}{15.6 \times 12.8}} = 0.818$ ✓ answer ✓ correct rounding

(b) $y = 5x^2 - \cos x$
 $y' = 10x + \sin x$ ✓✓ one for each

(c) A primitive of $x^3 + 5$
 is $\frac{x^4}{4} + 5x$ ✓✓ one for each

(d) $\frac{40^\circ}{360^\circ} = \frac{1}{9}$
 Area of a sector = $\frac{1}{9} \pi r^2$ ✓
 $= \frac{1}{9} \pi \times 9^2$ one for $40^\circ = \frac{\pi}{9}$
 $= 9\pi$ units ✓

(e) (i) $|x - 4| \leq 1$
 $3 \leq x \leq 5$ ✓



(f) $\frac{3}{3 - \sqrt{5}} \times \frac{3 + \sqrt{5}}{3 + \sqrt{5}}$ ✓
 $= \frac{9 + 3\sqrt{5}}{9 - 5}$ ✓

Question Two (12 marks)

(a) $P_n = P \left(1 + \frac{5}{100}\right)^n$
 $P_4 = 20000 \times (1.06)^4$ ✓
 $= \$25249.54$ ✓

(b) $y = e^{2x}$
 $y' = 2e^{2x}$ ✓

When $x = 0$, gradient = $2e^0 = 2$ ✓
 $y = e^0 = 1$

Eqn of tangent is $y - y_1 = m(x - x_1)$
 $y - 1 = 2(x - 0)$ ✓
 $y = 2x + 1$

(c) (i) midpoint of AC = $\left(\frac{9+6}{2}, \frac{6+3}{2}\right)$ ✓
 $= (7\frac{1}{2}, 1\frac{1}{2})$

(ii) midpoint of BD = $\left(\frac{0+15}{2}, \frac{4+1}{2}\right)$ ✓
 $= (7\frac{1}{2}, 1\frac{1}{2})$

(iii) $m_{BD} = \frac{4-1}{0-15} = -\frac{1}{3}$ ✓

(iv) $m_{AC} = \frac{6-3}{9-6} = 3$ ✓

$m_{AC} \times m_{BD} = 3 \times -\frac{1}{3} = -1$ ✓

$\therefore AC \perp BD$

(v) ABCD is rhombus ✓

(3)

● Question Three (12 marks)

● (a) (i) $y = \cos(2x+1)$

$$y' = -\sin(2x+1) \times 2$$

$$= -2\sin(2x+1) \quad \checkmark \checkmark$$

(ii) $y = \frac{x}{\log_e x}$

$$y' = \frac{\log_e x \times 1 - x \times \frac{1}{x}}{(\log_e x)^2}$$

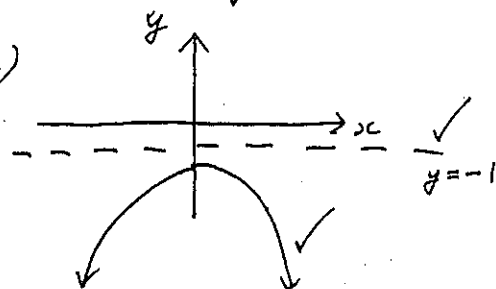
$$= \frac{\log_e x - 1}{(\log_e x)^2} \quad \checkmark \checkmark$$

(iii) $y = x^2 \tan x$

$$y' = x^2 \sec^2 x + 2x \tan x \quad \checkmark \checkmark$$

(b) $x^2 = -4(1)(y+2)$

$$V = (0, -2)$$



(c)

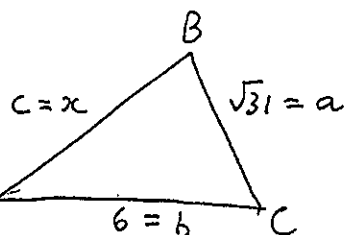
$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$31 = 36 + x^2 - 2 \times 6 \times x \cos 60^\circ$$

$$31 = 36 + x^2 - 2 \times 6 \times x \times \frac{1}{2}$$

$$31 = 36 + x^2 - 6x$$

$$x^2 - 6x + 5 = 0 \quad \checkmark$$



$$(x-5)(x-1) = 0$$

$$x = 5 \text{ or } x = 1 \quad \checkmark$$

Question 4 (12 marks)

(a) (i) $\int_0^{\ln 2} e^{2x} dx$

$$= \left[\frac{1}{2} e^{2x} \right]_0^{\ln 2}$$

$$= \frac{1}{2} e^{2 \ln 2} - \frac{1}{2} e^0$$

$$= \frac{1}{2} e^{\ln 4} - \frac{1}{2}$$

$$= 2 - \frac{1}{2} = 1\frac{1}{2} \quad \checkmark$$

(ii) $\int \frac{2x}{x^2+5} dx$

$$= \ln(x^2+5) + C \quad \checkmark$$

(d) $A = \int_0^\pi 1 + \cos x dx$

$$= \left[x + \sin x \right]_0^\pi \quad \checkmark$$

$$= \pi + \sin \pi - 0 - \sin 0$$

$$= \pi \text{ units}^2 \quad \checkmark$$

(b) $5 + 12 + 19 + \dots + 292$

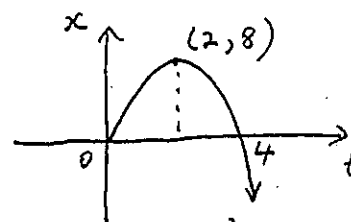
(i) # terms = $\frac{292-5}{7} + 1 = 42 \quad \checkmark$

(ii) $S_n = \frac{n}{2}(a+l)$

$$S_{42} = 21(5+292) \quad \checkmark$$

$$= 6237 \quad \checkmark$$

(c) (i)



$$x = 8t - 2t^2$$

$$= 2t(4-t)$$

$$V: t=2, x=8$$

✓ shape

✓ vertex

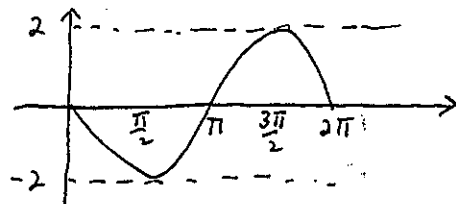
(ii) When $t=3$, $x = 24 - 18 = 6 \quad \checkmark$ (2 metr from

$$\text{Distance covered} = 8 + 2$$

$$= 10 \text{ metres} \quad \checkmark$$

Question 5 (12 marks)

(a) $y = -2 \sin x$ for $0 \leq x \leq 2\pi$



✓ shape and amplitude

✓ period.

(b) $l = r\theta$

$\frac{7}{12} = \theta$ ✓

$\theta = \left(\frac{7}{12} \times \frac{180}{\pi}\right)$ degrees

$\theta = 33^\circ$ ✓

(c) (i) $\left. \begin{array}{l} AC \text{ is common} \\ \angle BAC = \angle ACD \text{ (alt } \angle\text{'s, } AB \parallel CD) \\ \angle ABC = \angle ADC \text{ (given)} \end{array} \right\}$ ✓

$\therefore \triangle ABC \equiv \triangle CDA$ (AAS test) ✓

(ii) $\angle BCA = \angle DAC$ (matching sides of congruent Δ 's) ✓

But these are alternate angles
 $\therefore BC \parallel AD$ ✓

$\therefore ABCD$ is a parallelogram since two pairs of opposite sides are parallel ✓

(d) (i) $y = \sqrt{x-1}$

$y^2 = x-1$

$(y^2+1)^2 = x^2$ ✓

$V = \pi \int_0^3 y^4 + 2y^2 + 1 \, dy$ ✓

$= \pi \left[\frac{y^5}{5} + \frac{2y^3}{3} + y \right]_0^3$ ✓

348π ✓

Question 6 (12 marks)

(a) $\frac{dy}{dx} = x^{-\frac{1}{2}}$

$y = 2x^{\frac{1}{2}} + C$ ✓

When $x=4$ $-2 = 2\sqrt{4} + C$

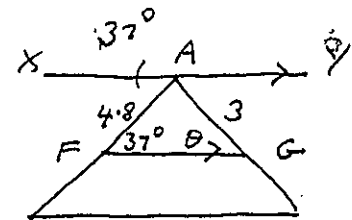
$y=-2$ $C = -6$

$y = 2\sqrt{x} - 6$ ✓

(b) (i) $\frac{x}{3} = \frac{8}{5}$ ✓

$x = \frac{24}{5}$

$xc = 4.8$ ✓



$\angle AFG = 37^\circ$ (alt $\angle$'s, $XY \parallel F$)

$\frac{\sin \theta}{4.8} = \frac{\sin 37^\circ}{3}$ ✓

$\theta = 74^\circ$ ✓

(c) $\log_a 400$

$= \log_a 25 + \log_a 16$

$= 2 \log_a 5 + 4 \log_a 2$

$= 2x + 4y$

✓ 'splitting logs'

✓ correct answer

(d).

(i)

x	0	1	2
y	1.732	2.172	3.064

✓ one correct

✓ three correct

(ii) $A \doteq \frac{h}{6} [f(a) + 4 \times f(\frac{a+b}{2}) + f(b)]$ ✓

$= \frac{2-0}{6} [1.732 + 4 \times 2.172 + 3.064]$

✓ (Follow through)

⑦

Question 7 (12 marks)

(a) (i) $y = x^4 - 4x^3 + 3$

$$y' = 4x^3 - 12x^2$$

$$y'' = 12x^2 - 24x = 12x(x-2)$$

Start pts where $y' = 0$

$$4x^3 - 12x^2 = 0$$

$$4x^2(x-3) = 0$$

$$x = 0 \text{ or } x = 3$$

When $x = 3$, $y'' = 108 - 72 > 0$

$\therefore$ Local minimum at $(3, -24)$

When $x = 0$, $y'' = 0$ (test inconclusive)

When $x = -1$, $y' = -16 < 0$

$x = 1$, $y' = -8 < 0$

$\therefore$ Horizontal point of inflexion at $(0, 0)$

(ii) Possible pts of inflexion where $y'' = 0$

$$12x^2 - 24x = 0$$

$$12x(x-2) = 0$$

$$x = 0 \text{ or } x = 2$$

At $x = 0$ there is a horizontal pt of inflexion

Consider $x = 1.9$, $y'' = 12(1.9)(-0.1) < 0$

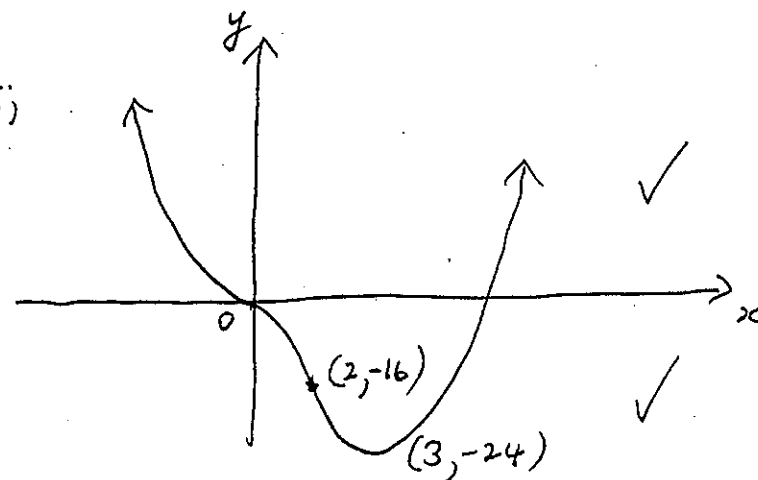
$x = 2.1$, $y'' = 12(2.1)(0.1) > 0$

Change in concavity

There is a point of inflexion at $(2, -16)$

⑧

(iii)



(b) Consider the quadratic

$$Kx^2 - 2x\sqrt{6} + (K+1)$$

This is positive definite when

$$a > 0 \text{ and } b^2 - 4ac < 0$$

i.e. $K > 0$ and $(2\sqrt{6})^2 - 4K(K+1) < 0$

$$24 - 4K^2 - 4K < 0$$

$$K^2 + K - 6 > 0$$

$$(K+3)(K-2) > 0$$

$K > 0$ and $K < -3$ or $K \geq 2$

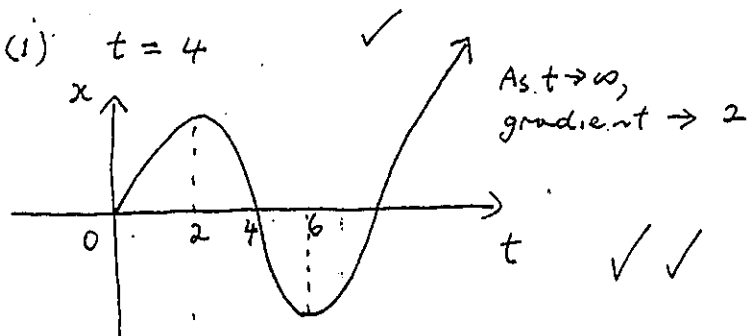
The quadratic is positive definite

When $K \geq 2$

Question 8 (12 marks)

(a) (i) $t = 4$ ✓

(ii)



(b) (i) $P_0 = 22000$ ✓
 $P = P_0 e^{kt}$
 $27000 = 22000 e^{5k}$ ✓
 $t = 5$
 $P = 27000$
 $k = \frac{1}{5} \ln \frac{27}{22}$ ✓
 $\approx 0.04095 \dots$

(ii) $P = 35000$ } $35000 = 22000 e^{kt}$ ✓
 $t = ?$ } $\frac{35}{22} = e^{kt}$
 $t = \frac{1}{k} \ln \frac{35}{22}$ ✓
 $= 11.3 \text{ years}$

(c) (i) Pts of intersection
 where $\sec x = 2$

$\cos x = \frac{1}{2}$
 $x = \frac{\pi}{3}$ ✓

(ii) $V = 2 \left\{ \int_0^{\frac{\pi}{3}} 2^2 dx - \int_0^{\frac{\pi}{3}} \sec^2 x dx \right\}$ ✓
 $= 2 \left\{ [4x]_0^{\frac{\pi}{3}} - [\tan x]_0^{\frac{\pi}{3}} \right\}$ ✓
 $= 2 \left\{ 4\pi - \sqrt{3} \right\} = (8\pi - 2\sqrt{3}) \text{ units}^2$

Question 9 (12 marks)

(a) $\frac{dV}{dt} = -2 - \frac{20}{t+1}$

(i) When $t = 0$, $\frac{dV}{dt} = -2 - \frac{20}{1}$ ✓
 $= -22 \text{ l/min.}$

(ii) $\frac{dV}{dt} = -2 - \frac{20}{t+1}$

$V = -2t - 20 \ln(t+1) + C$ ✓

When $t=0$ } $100 = 0 - 20 \ln 0 + C$
 $V=100$ } $C = 100$

$V = -2t - 20 \ln(t+1) + 100$ ✓

When $t = 5$, $V = -10 - 20 \ln 6 + 100$
 $\approx 54.2 \text{ litres}$ ✓

(b) (i) $A_1 = 250000 \times 1.005 - M$ ✓
 $A_2 = ((250000 \times 1.005) - M) 1.005 - M$
 $= 250000 \times (1.005)^2 - 1.005M - M$
 $= 250000 \times (1.005)^2 - M(1 + 1.005)$ ✓
 $A_3 = 250000 \times (1.005)^3 - M(1 + 1.005 + 1.005^2)$

(ii) Loan is paid off over 240 months = 20y

(iii) $0 = 250000 \times (1.005)^{240} - M(1 + 1.005 + \dots + 1.005^{239})$
 $250000 \times (1.005)^{240} = M \left[\frac{a(r^n - 1)}{r - 1} \right]$

$$250000 \times (1.005)^{240} = M \left(\frac{(1.005)^{240} - 1}{1.005 - 1} \right) \quad (11)$$

$$M = \$1791.08$$

(IV) Suppose $M = \$2000$ we need to find n so that

$$250000 \times (1.005)^n = 2000 \left(\frac{(1.005)^n - 1}{0.005} \right)$$

$$250000 \times 0.005 \times (1.005)^n = 2000 \times 1.005^n - 2000$$

$$1250 \times (1.005)^n = 2000 \times 1.005^n - 2000$$

$$(1.005)^n (2000 - 1250) = 2000$$

$$(1.005)^n = \frac{8}{3}$$

$$n \ln(1.005) = \ln \frac{8}{3}$$

$$n = \frac{\ln \frac{8}{3}}{\ln 1.005}$$

$$= 196.65 \text{ months}$$

Loan is paid off approximately 43 months earlier.

Question 10

(a) (i) $y = x^{\frac{2}{3}}$
 $y' = \frac{2}{3} x^{-\frac{1}{3}}$
 $y'' = -\frac{2}{9} x^{-\frac{4}{3}}$
 $y''' = -\frac{2}{9} \cdot 3 \sqrt[3]{\frac{1}{x^4}}$

$$x^4 > 0 \text{ for } x \neq 0$$

$$\therefore y'' < 0 \text{ for } x \neq 0$$

(ii) Solve $x^{\frac{2}{3}} = \frac{x}{2}$

Cube both sides

$$x^2 = \frac{x^3}{8}$$

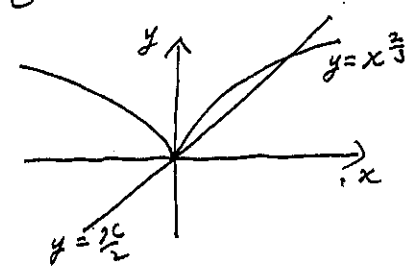
$$x^3 - 8x^2 = 0$$

$$x^2(x - 8) = 0$$

$$x = 0 \text{ or } x = 8$$

Solution to $x^{\frac{2}{3}} \leq \frac{x}{2}$

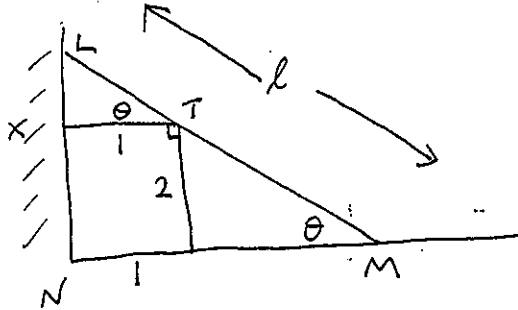
$$\text{is } x \geq 8 \text{ or } x = 0 \quad \checkmark$$



Give one mark for sketch OR

Question 10 (12 marks) cont'd

(b)



- (i) Draw XT parallel to NM so that $\angle LTX = \theta$.

$$\frac{TL}{TX} = \sec \theta \quad \checkmark \quad \frac{TM}{2} = \csc \theta$$

$$TL = \sec \theta \quad TM = 2 \csc \theta$$

$$LM = \sec \theta + 2 \csc \theta$$

$$l = \frac{1}{\cos \theta} + \frac{2}{\sin \theta} \quad \checkmark$$

$$(ii) \quad l = (\cos \theta)^{-1} + 2(\sin \theta)^{-1}$$

$$l' = -\frac{-\sin \theta}{\cos^2 \theta} - 2\frac{\cos \theta}{\sin^2 \theta} \quad \checkmark$$

Start pts where $l' = 0$

$$\frac{\sin \theta}{\cos^2 \theta} = \frac{2 \cos \theta}{\sin^2 \theta} \quad \checkmark$$

$$\frac{\sin^3 \theta}{\cos^3 \theta} = 2$$

$$\tan^3 \theta = 2 \quad \checkmark$$

$$\tan \theta = \sqrt[3]{2}$$

- (iii) Minimum l at $\theta = \tan^{-1} \sqrt[3]{2}$

or end points $\theta = 55^\circ$ or $\theta = 70^\circ$

$$\theta = \tan^{-1} \sqrt[3]{2} \quad \checkmark$$

$$\hat{=} 51^\circ 34' \quad \text{which is outside the safety constraint.}$$

$$\text{Test } \theta = 55^\circ \quad l = \frac{2}{\sin 55^\circ} + \frac{1}{\cos 55^\circ} \hat{=} 4.18$$

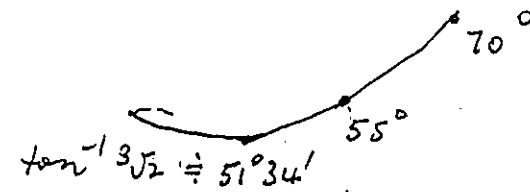
$$\checkmark \quad \theta = 70^\circ \quad l = \frac{2}{\sin 70^\circ} + \frac{1}{\cos 70^\circ} \hat{=} 5.05$$

So minimum l when $\theta = 55^\circ \quad \checkmark$
and the length is 4.18 m

OR You can show that l is increasing for $\theta > \tan^{-1} \sqrt[3]{2}$

θ	50°	$\tan^{-1} \sqrt[3]{2}$	52°
l'	-	0	+

So the curve looks like



Clearly minimum at $\theta = 55^\circ$